

## SEMESTER S3

### DATA STRUCTURES AND ALGORITHMS

(Common to CS/CA/CM/CD/CR/AI/AM/AD/CB/CN/CC/CU/CI/CG)

|                                            |                 |                    |                |
|--------------------------------------------|-----------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>PCCST303</b> | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:1:0:0         | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 4               | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | UCEST105        | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. To provide the learner a comprehensive understanding of data structures and algorithms.
2. To prepare them for advanced studies or professional work in computer science and related fields.

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                  | <b>Contact Hours</b> |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | <b>Basic Concepts of Data Structures</b><br>Definitions; Data Abstraction; Performance Analysis - Time & Space Complexity, Asymptotic Notations; Polynomial representation using Arrays, Sparse matrix ( <i>Tuple representation</i> ); Stacks and Queues - Stacks, Multi-Stacks, Queues, Circular Queues, Double Ended Queues; Evaluation of Expressions- Infix to Postfix, Evaluating Postfix Expressions. | <b>11</b>            |
| <b>2</b>          | <b>Linked List and Memory Management</b><br>Singly Linked List - Operations on Linked List, Stacks and Queues using Linked List, Polynomial representation using Linked List; Doubly Linked List; Circular Linked List; Memory allocation - First-fit, Best-fit, and Worst-fit allocation schemes; Garbage collection and compaction.                                                                        | <b>11</b>            |
| <b>3</b>          | <b>Trees and Graphs</b><br>Trees :- Representation Of Trees; Binary Trees - Types and Properties, Binary Tree Representation, Tree Operations, Tree Traversals; Expression Trees; Binary Search Trees - Binary Search Tree Operations; Binary Heaps - Binary Heap Operations, Priority Queue.<br>Graphs :- Definitions; Representation of Graphs; Depth First Search and                                     | <b>11</b>            |

|          |                                                                                                                                                                                                                                                                                                                                                                    |           |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|          | Breadth First Search; Applications of Graphs - Single Source All Destination.                                                                                                                                                                                                                                                                                      |           |
| <b>4</b> | <b>Sorting and Searching</b><br>Sorting Techniques :- Selection Sort, Insertion Sort, Quick Sort, Merge Sort, Heap Sort, Radix Sort.<br>Searching Techniques :- Linear Search, Binary Search, Hashing - Hashing functions : Mid square, Division, Folding, Digit Analysis; Collision Resolution : Linear probing, Quadratic Probing, Double hashing, Open hashing. | <b>11</b> |

**Course Assessment Method**  
**(CIE: 40 marks, ESE: 60 marks)**

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Assignment/<br/>Microproject</b> | <b>Internal<br/>Examination-1<br/>(Written)</b> | <b>Internal<br/>Examination- 2<br/>(Written)</b> | <b>Total</b> |
|-------------------|-------------------------------------|-------------------------------------------------|--------------------------------------------------|--------------|
| <b>5</b>          | <b>15</b>                           | <b>10</b>                                       | <b>10</b>                                        | <b>40</b>    |

**End Semester Examination Marks (ESE)**

*In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions*

| <b>Part A</b>                                                                                                                                                                                  | <b>Part B</b>                                                                                                                                                                                                                                                                                                | <b>Total</b> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <ul style="list-style-type: none"> <li>2 Questions from each module.</li> <li>Total of 8 Questions, each carrying 3 marks</li> </ul> <p style="text-align: center;"><b>(8x3 =24 marks)</b></p> | <ul style="list-style-type: none"> <li>Each question carries 9 marks.</li> <li>Two questions will be given from each module, out of which 1 question should be answered.</li> <li>Each question can have a maximum of 3 sub divisions.</li> </ul> <p style="text-align: center;"><b>(4x9 = 36 marks)</b></p> | <b>60</b>    |

### Course Outcomes (COs)

At the end of the course students should be able to:

| Course Outcome |                                                                                                 | Bloom's Knowledge Level (KL) |
|----------------|-------------------------------------------------------------------------------------------------|------------------------------|
| <b>CO1</b>     | Identify appropriate data structures for solving real world problems.                           | <b>K3</b>                    |
| <b>CO2</b>     | Describe and implement linear data structures such as arrays, linked lists, stacks, and queues. | <b>K3</b>                    |
| <b>CO3</b>     | Describe and Implement non linear data structures such as trees and graphs.                     | <b>K3</b>                    |
| <b>CO4</b>     | Select appropriate searching and sorting algorithms to be used in specific circumstances.       | <b>K3</b>                    |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

### CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 3   | 3   | 3   |     |     |     |     |     |     |      |      | 3    |
| <b>CO2</b> | 3   | 3   | 3   |     |     |     |     |     |     |      |      | 3    |
| <b>CO3</b> | 3   | 3   | 3   |     |     |     |     |     |     |      |      | 3    |
| <b>CO4</b> | 3   | 3   | 3   |     |     |     |     |     |     |      |      | 3    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| Text Books |                                      |                                                                     |                       |                  |
|------------|--------------------------------------|---------------------------------------------------------------------|-----------------------|------------------|
| Sl. No     | Title of the Book                    | Name of the Author/s                                                | Name of the Publisher | Edition and Year |
| 1          | Fundamentals of Data Structures in C | Ellis Horowitz, Sartaj Sahni and Susan Anderson-Freed,              | Universities press,   | 2/e, 2007        |
| 2          | Introduction to Algorithms           | Thomas H Cormen, Charles Leiserson, Ronald L Rivest, Clifford Stein | PHI                   | 3/e, 2009        |

| Reference Books |                                                   |                                            |                       |                  |
|-----------------|---------------------------------------------------|--------------------------------------------|-----------------------|------------------|
| Sl. No          | Title of the Book                                 | Name of the Author/s                       | Name of the Publisher | Edition and Year |
| 1               | Classic Data Structures                           | Samanta D.                                 | Prentice Hall India.  | 2/e, 2018        |
| 2               | Data Structures and Algorithms                    | Aho A. V., J. E. Hopcroft and J. D. Ullman | Pearson Publication.  | 1/e, 2003        |
| 3               | Introduction to Data Structures with Applications | Tremblay J. P. and P. G. Sorenson          | Tata McGraw Hill.     | 2/e, 2017        |
| 4               | Theory and Problems of Data Structures            | Lipschuts S.                               | Schaum's Series       | 2/e, 2014        |

| Video Links (NPTEL, SWAYAM...) |                                                                                                                                                               |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module No.                     | Link ID                                                                                                                                                       |
| 1                              | <a href="https://nptel.ac.in/courses/106102064">https://nptel.ac.in/courses/106102064</a>                                                                     |
| 2                              | <a href="https://ocw.mit.edu/courses/6-851-advanced-data-structures-spring-2012/">https://ocw.mit.edu/courses/6-851-advanced-data-structures-spring-2012/</a> |